

What is Data Science?

Data science, as its name suggests, is all about the data. Hence, we can define it as, "**A field of deep study of data that includes extracting useful insights from the data, and processing that information using different tools, statistical models, and Machine learning algorithms.**" It is a concept that is used to handle big data that includes data cleaning, data preparation, data analysis, and data visualization.

A data scientist collects the raw data from various sources, prepares and pre-processes the data, and applies machine learning algorithms, predictive analysis to extract useful insights from the collected data.

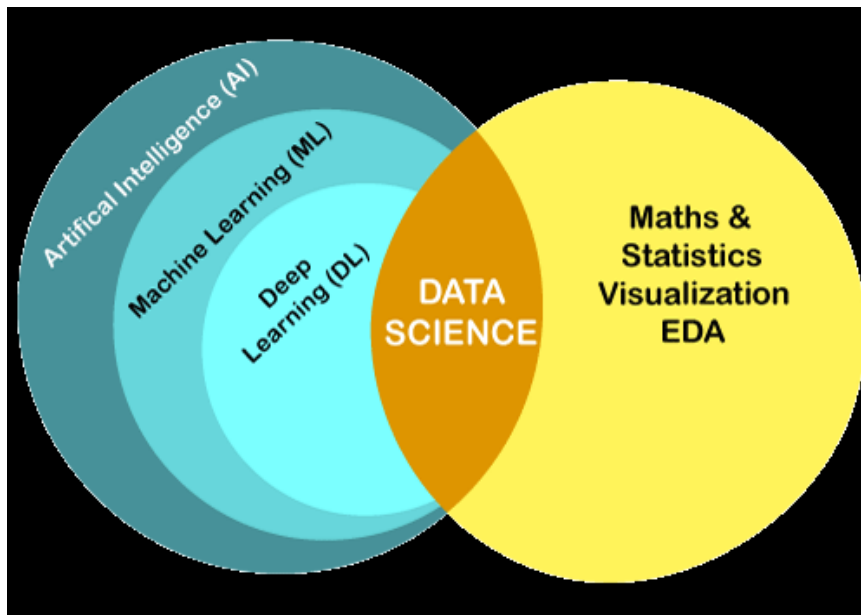
For example, Netflix uses data science techniques to understand user interest by mining the data and viewing patterns of its users.

Skills Required to become Data Scientist

- o An excellent programming knowledge of **Python, R, SAS, or Scala.**
- o Experience in SQL database Coding.
- o Knowledge of Machine Learning Algorithms.
- o Deep Knowledge of Statistics concepts.
- o Data Mining, cleaning, and Visualizing skills.
- o Skills to use Big data tools such as Hadoop.

Data Science is the study of *data cleansing, preparation, and analysis*, while machine learning is a branch of AI and subfield of data science. Data Science and Machine Learning are the two popular modern technologies, and they are growing with an immoderate rate. But these two buzzwords, along with artificial intelligence and deep learning are very confusing term, so it is important to understand how they are different from each other. In this topic, we will understand the difference between Data Science and Machine Learning only, and how they relate to each other.

Data Science and **Machine Learning** are closely related to each other but have different functionalities and different goals. At a glance, **Data Science** is a field to study the approaches to find insights from the raw data. Whereas, Machine Learning is a technique used by the group of data scientists to enable the machines to learn automatically from the past data. To understand the difference in-depth, let's first have a brief introduction to these two technologies.



life cycle of Data Science. The different steps that occur in Data science lifecycle are as follows:



1. **Business Requirements:** In this step, we try to understand the requirement for the business problem for which we want to use it.

Suppose we want to create a recommendation system, and the business requirement is to increase sales.

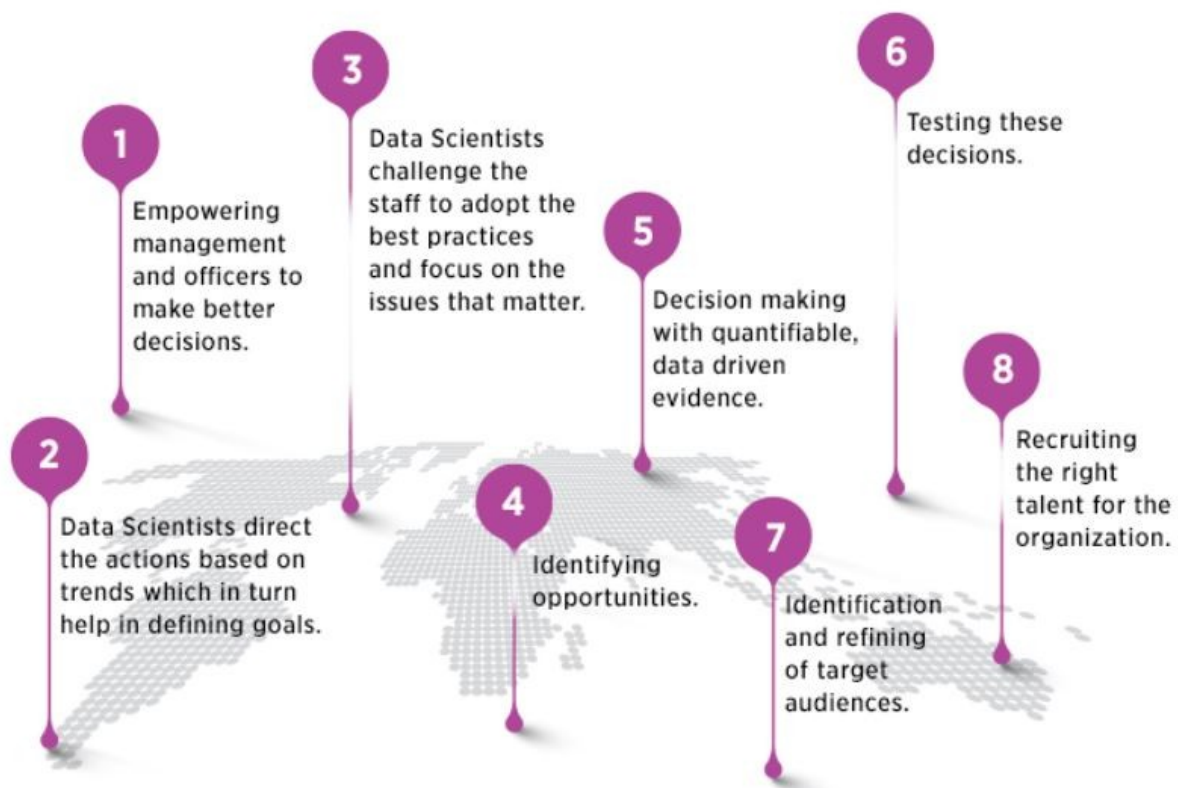
2. **Data Acquisition:** In this step, the data is acquired to solve the given problem. For the recommendation system, we can get the ratings provided by the user for different products, comments, purchase history, etc.
3. **Data Processing:** In this step, the raw data acquired from the previous step is transformed into a suitable format, so that it can be easily used by the further steps.
4. **Data Exploration:** It is a step where we understand the patterns of the data, and try to find out the useful insights from the data.
5. **Modeling:** The data modeling is a step where machine learning algorithms are used. So, this step includes the whole machine learning process. The machine learning process involves importing the data, data cleaning, building a model, training the model, testing the model, and improving the model's efficiency.
6. **Deployment & Optimization:** This is the last step where the model is deployed on an actual project, and the performance of the model is checked.

Data science has become an essential part of any industry today. It's a method for transforming business data into assets that help organizations improve revenue, reduce costs, seize business opportunities, improve customer experience, and more. Data science is one of the most debated topics in the industries these days. Its popularity has grown over the years, and companies have started implementing data science techniques to grow their business and increase customer satisfaction. Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information, and make business decisions.



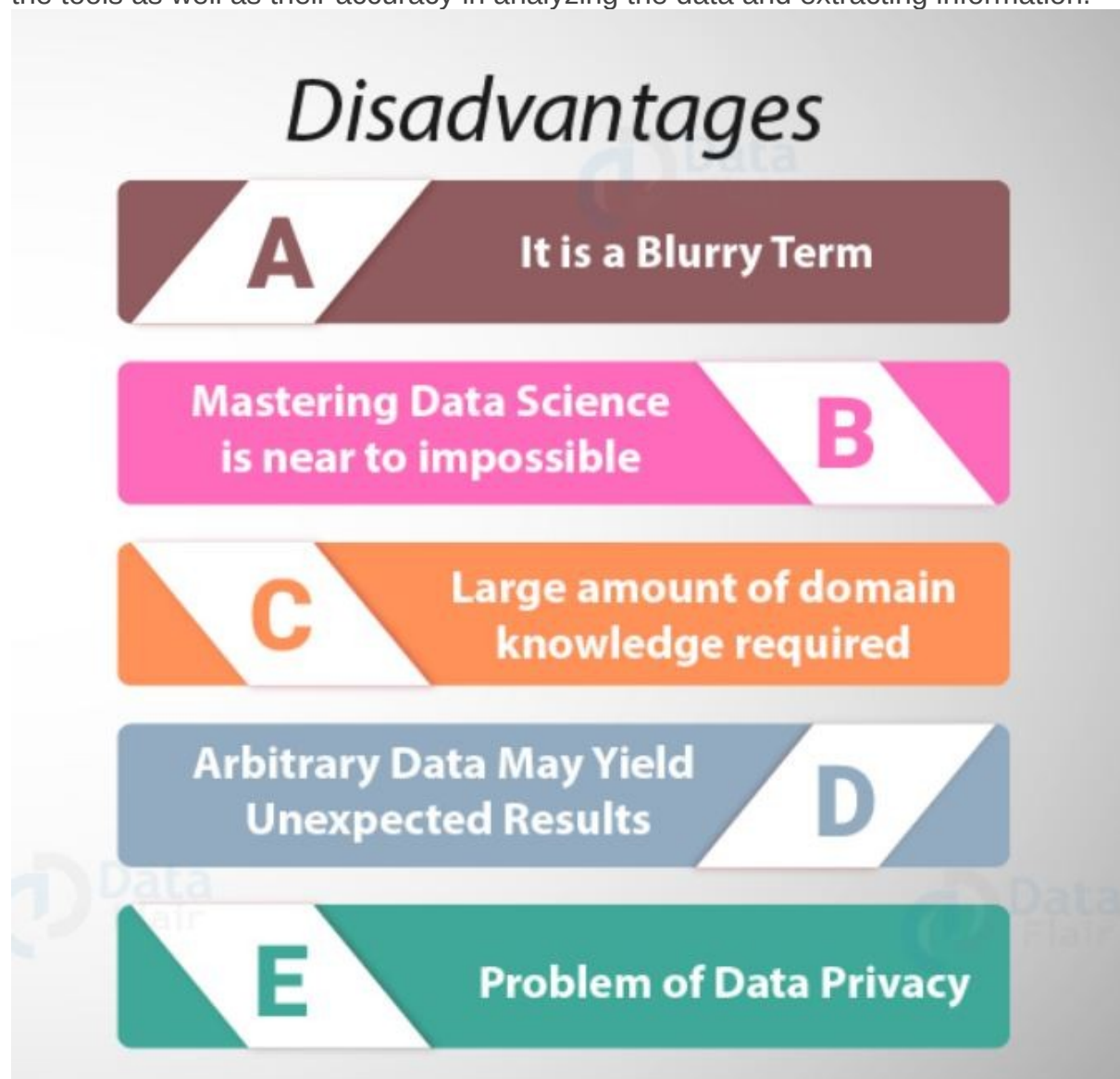
Advantages of Data Science :- In today's world, data is being generated at an alarming rate. Every second, lots of data is generated; be it from the users of Facebook or any other social networking site, or from the calls that one makes, or the data which is being generated from different organizations. And because of this huge amount of data the value of the field of Data Science has a number of advantages. Some of the advantages are mentioned below :-

- **Multiple Job Options :-** Being in demand, it has given rise to a large number of career opportunities in its various fields. Some of them are *Data Scientist, Data Analyst, Research Analyst, Business Analyst, Analytics Manager, Big Data Engineer, etc.*
- **Business benefits :-** Data Science helps organizations knowing how and when their products sell best and that's why the products are delivered always to the right place and right time. Faster and better decisions are taken by the organization to improve efficiency and earn higher profits.
- **Highly Paid jobs & career opportunities :-** As Data Scientist continues being the sexiest job and the salaries for this position are also grand. According to a Dice Salary Survey, the annual average salary of a Data Scientist \$106,000 per year.
- **Hiring benefits :-** It has made it comparatively easier to sort data and look for best of candidates for an organization. Big Data and data mining have made processing and selection of CVs, aptitude tests and games easier for the recruitment teams.



Disadvantages of Data Science :- Everything that comes with a number of benefits also has some consequences . So let's have a look at some of the disadvantages of Data Science :-

- **Data Privacy :-** Data is the core component that can increase the productivity and the revenue of industry by making game-changing business decisions. But the information or the insights obtained from the data can be misused against any organization or a group of people or any committee etc. Extracted information from the structured as well as unstructured data for further use can also misused against a group of people of a country or some committee.
- **Cost :-** The tools used for data science and analytics can cost a lot to an organization as some of the tools are complex and require the people to undergo a training in order to use them. Also, it is very difficult to select the right tools according to the circumstances because their selection is based on the proper knowledge of the tools as well as their accuracy in analyzing the data and extracting information.



Top 9 best tools which we use in Data Science :- It is required that they have a clear understanding of the tools that are necessary for the programming to work. we decided to provide a little insight into the tools that can be used for data visualization, statistical programming languages, algorithms, and databases. These tools will help speed up your process as you do not have to further search anywhere else for what you need.

1. **DataRobot** :- It is a global automated Machine Learning platform. With the capabilities of Data Science, Machine Learning, Statistical Modeling, Artificial Intelligence, Augmented Analytics, Machine Learning Operations (MLOps), Time Series Modeling.
2. **MLBASE** :- One of the best Data Science tools and provides distributed and statistical techniques that are key to transforming big data into actionable knowledge. It provides functionality to end-users for a wide variety of standard machine learning tasks such as classification, regression, collaborative filtering, and more general exploratory data analysis techniques
3. **Apache Graph** :- Apache Graph supports high-level scalability. It is an iterative graph processing system that has been specially developed for this purpose. This was derived from the Pregel model but comes with more number of features and functionalities when compared with the Pregel model. This open-source model helps data scientists to utilize the underlying potential of structured datasets at a large scale.
4. **Apache Spark** :- This is another free tool that offers cluster computing in a blink of the eye, which is at lightning bolt speed. Today, a number of organizations are using Spark for processing large datasets. This data scientist tool is capable of accessing diverse data sources, which include HDFS, HBase, S3, and Cassandra.
5. **Cascading** :- It is specifically for data scientists who are building big data apps on Apache Hadoop. It allows users to solve both complex and simple data problems, using cascading. This is because it offers computation engines, data processing, scheduling capabilities, and systems integration framework.
6. **TABLEAU** :- It is a Data Science visualization software with powerful graphics to make interactive visualizations. It can interface with databases, spreadsheets, OLAP (Online Analytical Processing) cubes. It provides the capability of visualizing the geographical data and for plotting longitudes and latitudes in maps.
7. **TENSORFLOW** :- This is an ML tool, which is widely used for advanced Machine Learning algorithms like Deep Learning. It is an open-source and ever-evolving toolkit which is known for its performance and high computational abilities.
8. **SAP HANA** :- It is an effective tool from SAP with SAP HANA Predictive Analysis Library (PAL).
9. **MANGODB** :- This is another Data Analysis tool that is quite popular since it allows cross-platform document orientation. It has a basic query and aggregation framework, but to do more advanced analytics. It is a perfect choice to iterate ML training experiments.

Conclusion :- Everything in this world has its pros and cons, but we should not neglect the fact that our work gets easier when we use such tools that helps us in not only extracting the information but also in reducing the development time and cost of

the product to be delivered. Data Science uses the data (Big Data) to make some decisions that can add increase the profit to any business in a very effective way. After going through the pros and cons of Data Science we can now have a better thought of it at a larger picture. Even after having a lot of advantages and being a very interesting and exciting field it has a few disadvantages also. Considering both the sides will help you to decide that, *"Whether you want to make us of Data Science or not?* And will help you in making a very important career decision.